Roll Number												

M(CSE)301 - DISCRETE MATHEMATICS

TIME ALLOTTED: 3Hours **FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)
Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. A simple graph has (1) CO1 BL1
a) no parallel edges
b) no loops
c) no parallel edges and no loops
d) none

2. $(p \rightarrow \sim q) \rightarrow p$ is a contradiction (1) CO1 BL1
a) True
b) False

3. The proposition p∧(~pvq) is (1) CO1 BL1

a) Tautology
b) Logical equivalence to p∧q
c) Logical equivalence to pvq
d) Contradiction

4. The number of edges in a complete graph with n vertices is (1) CO1 BL1
a) n
b) n(n-1)/2
c) n(n+1)/2
d) n/2

5. The minimum number of pedant vertices in a tree with five vertices is (1) CO1 BL1
a) 1
b) 2
c) 3
d) none of these

6. In the group {1, -1, i, -i} under multiplication, order of -i is (1) CO1 BL1
a) 1
b) 2
c) 4
d) none of these

7. The symmetric group S_3 has (1) CO1 BL1
a) 6 elements
b) 8 elements
c) 9 elements
d) 12 elements

8. The only generators of the cyclic group (Z, +) is (1) CO1 BL1
a) 1
b) 0,1
c) 1,-1

d) All positive integers

9. Product of any 3 consecutive integers is divisible by (1) CO1 BL1
a) 6
b) 7
c) 4
d) none of these

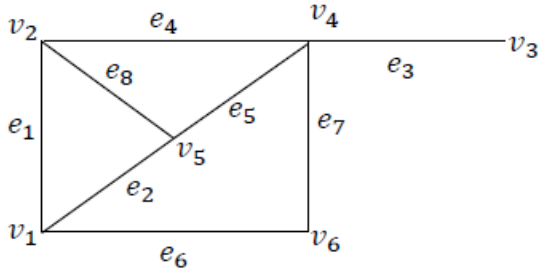
10. If a is prime to b, then a^2+b^2 is prime to (1) CO1 BL1
a) a^2b
b) a^2b^2
c) ab
d) a+b

11. Which of the following statement is a proposition? (1) CO1 BL1
a) Get me a glass of water
b) Best wishes!
c) The only even prime number is 2.
d) none of these

12. A Poset in which every pair of elements has both a least upper bound and a greatest lower bound is termed as _____ (1) CO1 BL1
a) lattice
b) sublattice
c) walk
d) trail

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

13. Determine the adjacent matrix of the following graph (5) CO2 BL2



14. Show that the group $(\mathbb{Z}_5, +)$, i.e. the additive group of all integers modulo 5 is cyclic . Find allgenerators of $\mathbb{Z}_5$. (5) CO4 BL4

15. Answer the questions given below (5)

a) If gcd(a,b)=1,then then prove that gcd (a+b, a-b)=1 or 2. (3) CO3 BL3

b) Find the remainder when the sum 1!+2!+3!+4!+…..+100! is divided by 30. (2) CO4 BL4

16. Write DNF of the following statement $p\rightarrow(p\wedge(q\rightarrow p))$ (5) CO2 BL2

17. A relation R is defined on the set Z by "aRb if and only if a-b is divisible by 5" for a, b∈Z.Examine if R is an equivalence relation on Z. (5) CO4 BL4

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

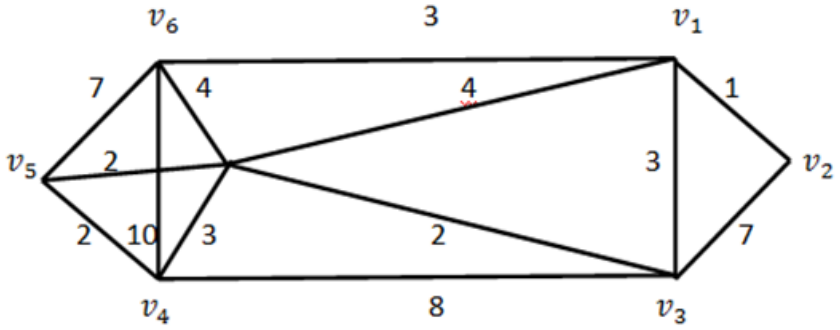
18a. Solve the recurrence relation using the generating function (7) CO5 BL5

$$a_{n+2} - 5a_{n+1} + 6a_n = 2,\ n \geq 0,\ a_0 = 3,\ a_1 = 7.$$

18b. Let A ={1, 2, 3, 4, 6, 8} be ordered by the relation "a divides b". Then find it's Hasse diagram. (5) CO3 BL3

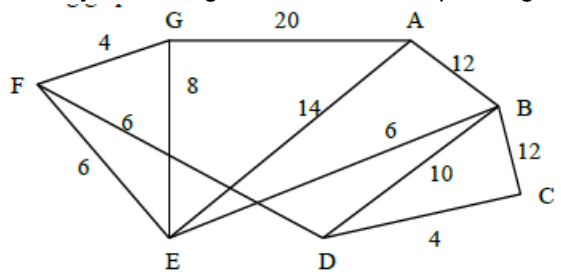
18c. Use theory of congruence to prove that n≥1, 17 divides $(2^{3n+1} + 3.5^{2n+1})$ (3) CO2 BL2

19a. (6) CO5 BL5



Applying Dijkstra's algorithm find the shortest path from v_2 to v_5.

19b. Find by Prim's algorithm a minimal spanning tree from the following path (9) CO3 BL3



20a. Show that all roots of the equation $x^4=1$ forms a commutative group under the operation usual multiplication. (5) CO3 BL3

20b. If y be an element of a group (G, ∗) and oder of y is 20, find the order of y^8. (5) CO3 BL3

20c. Let A and B be two permutations.Then show that AB is not equal to BA.Where A and B are given by (5) CO2 BL2

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}.$$

21a. Using truth table show that p→(q∨r)≡(p→q)∨(q→r). (5) CO2 BL2

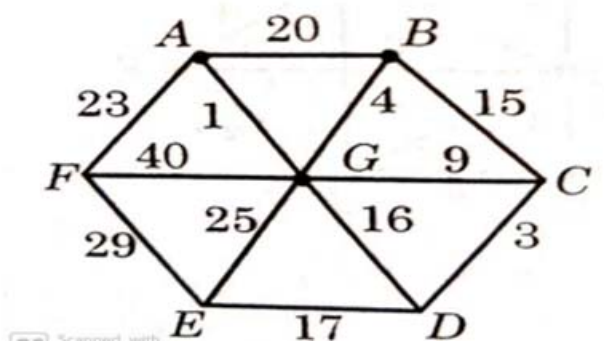
21b. Check whether (p∧q)↔ (q∧r)∨(r∧~p) is a tautology or not. (5) CO2 BL2

21c. Verify the following result $3^{302} \equiv 4 \ mod(5)$ (5) CO4 BL4

22a. Define regular graph. Draw a graph or di-graph accordingly corresponding to the following adjacency matrix (5) CO3 BL3

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

22b. By Kruskal's Algorithm find a minimal spanning tree and corresponding weight in the following graph (5) CO4 BL4



22c. Calculate the gcd (1761, 189) and express it as 1761u+189v (5) CO4 BL4